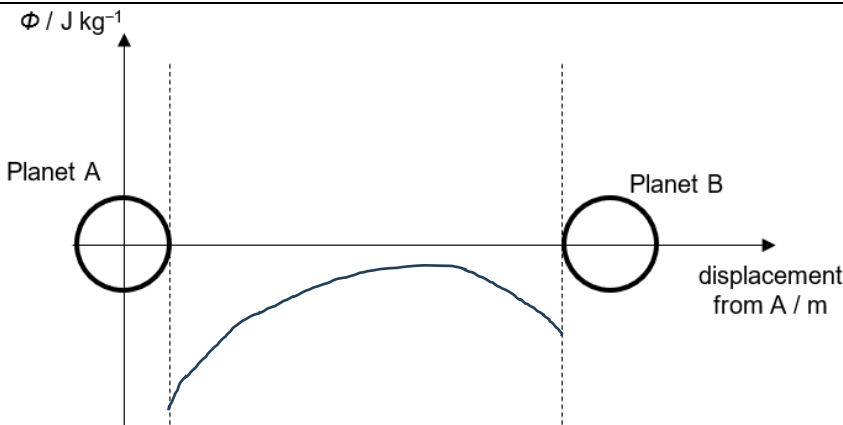
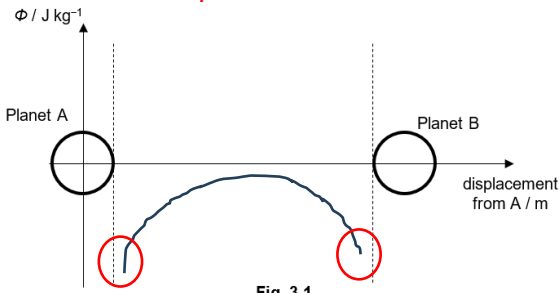
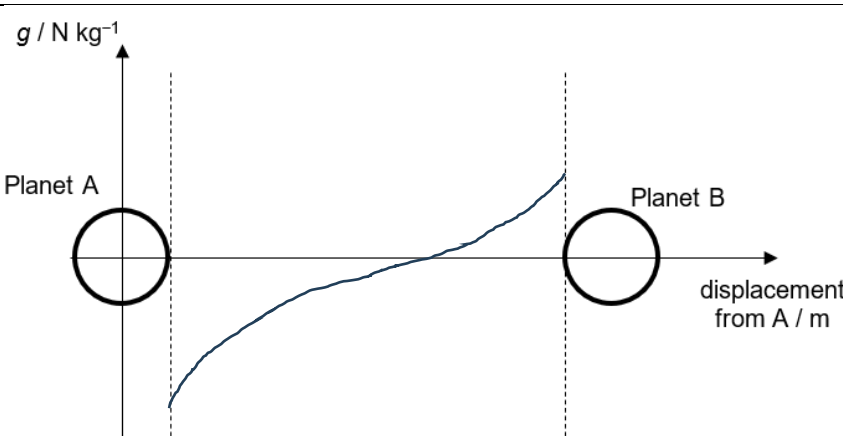


			$= (2.10 - 0) / (0.000)$ $= 36 \text{ N}$	A1
3	(a)	(i)	The positive direction was defined to be the direction from A to B. $\text{gravitational field strength} = -\frac{GM_A}{r^2} + \frac{GM_B}{r^2}$ $= \frac{6.67 \times 10^{-11}}{(0.5 \times 3.85 \times 10^8)^2} (-5.07 \times 10^{24} + 3.23 \times 10^{24})$ $= -3.31 \times 10^{-3} \text{ N kg}^{-1}$ Accept $3.31 \times 10^{-3} \text{ N kg}^{-1}$	M1 A1

		(ii)  Fig. 3.1 General shape correct. Gravitational potential at surface of A is lower than that at B Marker's comments: - Many students left a gap between the line to the dotted lines (the surface of both planets). This is not acceptable since the instruction stated specifically to draw from the surface of one planet to the other. Marks cannot be awarded for incomplete diagrams. - Some students drew the curve as if the dotted lines are the asymptotes. This is incorrect, since it suggests that the magnitude of the gravitational potential at the surface of the planet approaches infinity. Incorrect example:  Fig. 3.1	B1
		(iii)  Fig. 3.2 General shape	B1 B1

			Magnitude of g at surface of A larger than that at B and neutral point closer to planet B Marker's comments: Similar to previous question, some students drew the curve as if the dotted lines are the asymptotes. This is incorrect, since it suggests that the magnitude of the gravitational field strength at the surface of the planet approaches infinity.	
	(b)	(i)	- By Newton's 3rd law of motion, Force on C by D = force on D by C - The same magnitude of force provides for the centripetal force on each star about P. - The angular velocity ω is the same for both stars. Hence, $M_C r_C \omega^2 = M_D r_D \omega^2$ $\frac{M_D}{M_C} = \frac{r_C}{r_D}$ $\therefore \frac{r_C}{r_D} = \frac{1}{2}$	M1
		(ii)	Since $\frac{r_C}{r_D} = \frac{1}{2}$ and $\frac{M_C}{M_D} = 2$, $\text{gravitational force} = \frac{GM_C \frac{M_C}{2}}{(r_C + 2r_C)^2} = \frac{GM_C^2}{18r_C^2}$	M1
		(iii)	Gravitational force on C by D provides for centripetal force on C $\frac{GM_C^2}{18r_C^2} = M_C r_C \omega^2$ $M_C = \frac{18r_C^3}{G} \left(\frac{2\pi}{T} \right)^2$ $= \frac{18(2.40 \times 10^{12})^3}{6.67 \times 10^{-11}} \left(\frac{2\pi}{3.84 \times 10^9} \right)^2$ $= 9.99 \times 10^{30} \text{ kg}$	M1 M1 A1
4	(c)	(i)	... f3	M1